

Year 9 Science – Booklet 8: Biology

Genetics & DNA — ANSWERS

Quick Test (p.3)

1. In the nucleus of the cell, carried on the chromosomes.
2. 46 (23 pairs).
3. 23.
4. 20.
5. bb (black hair is recessive, so a person showing it must have two copies of the recessive allele).

Section A – Multiple Choice (5 marks)

1. a) on a chromosome
2. d) 2
3. c) radiation
4. c) a disease of the red blood cells
5. b) 1

Section B (13 marks)

1. Fill in the gaps (8 marks)

Inside a cell is the nucleus, which controls all our inherited features. Inside the nucleus are thread-like **chromosomes**. On the chromosomes are **pairs of genes**. Humans have 46 chromosomes. The genes are made up of a chemical called **DNA**. We inherit two genes for every feature. Genes can be more powerful and called **dominant** or less powerful and called **recessive**. Occasionally during the inheritance process things can go wrong. This is called a **mutation**. The chances of this happening are increased by exposure to certain chemicals and **radiation**.

2. Red rose × white rose → all red offspring

- a). Red (since crossing red with white gave only red offspring, red must be dominant).
- b). Rr (heterozygous – one dominant R from the red parent, one recessive r from the white parent).
- c) Punnet square for Rr × Rr:

	R	r
R	RR	Rr
r	Rr	rr

Offspring: RR, Rr, Rr, rr → 3 red (RR, Rr, Rr) : 1 white (rr), matching the observed "three times as many red roses as white roses".

Section C (7 marks)

1. Genes and appearance

- a). The environment (e.g. diet, lifestyle) also affects what we look like, not just our genes.
- b). The nucleus.
- c). 46 (23 pairs).
- d). 20.

2. Punnet square (mother bb × father Bb): offspring Bb , bb , Bb , bb

- a). 2 out of 4 (50%) brown-eyed children.
- b). 2 out of 4 (50%) blue-eyed children.
- c). Bb (heterozygous – brown eyes, but carrying one recessive blue allele).